Self-Supervised Learning for Crack Detection

SimCLR

BYOL

MAE

Problem Context

~504 Images
(Small Dataset)

60% Crack / 40%
No Crack

Fine Crack
Structures

Lighting Variation

Problem: Overfitting
& Weak
Generalization

Why Self-Supervised Learning?

- Learns from data without labels
- Improves feature learning
- Works well with small datasets
- Better generalization

SimCLR (Contrastive Learning)

1

Compare two views of same image

2

Pull similar together

3

Push different apart

4

Best for texture & edge learning

BYOL (No Negative Pairs)

- Learns from itself
- Online & target networks
- Stable training
- Best for small datasets

MAE (Masked Autoencoder)

1

Hide parts of image

2

Reconstruct missing patches

3

Best for structural understanding

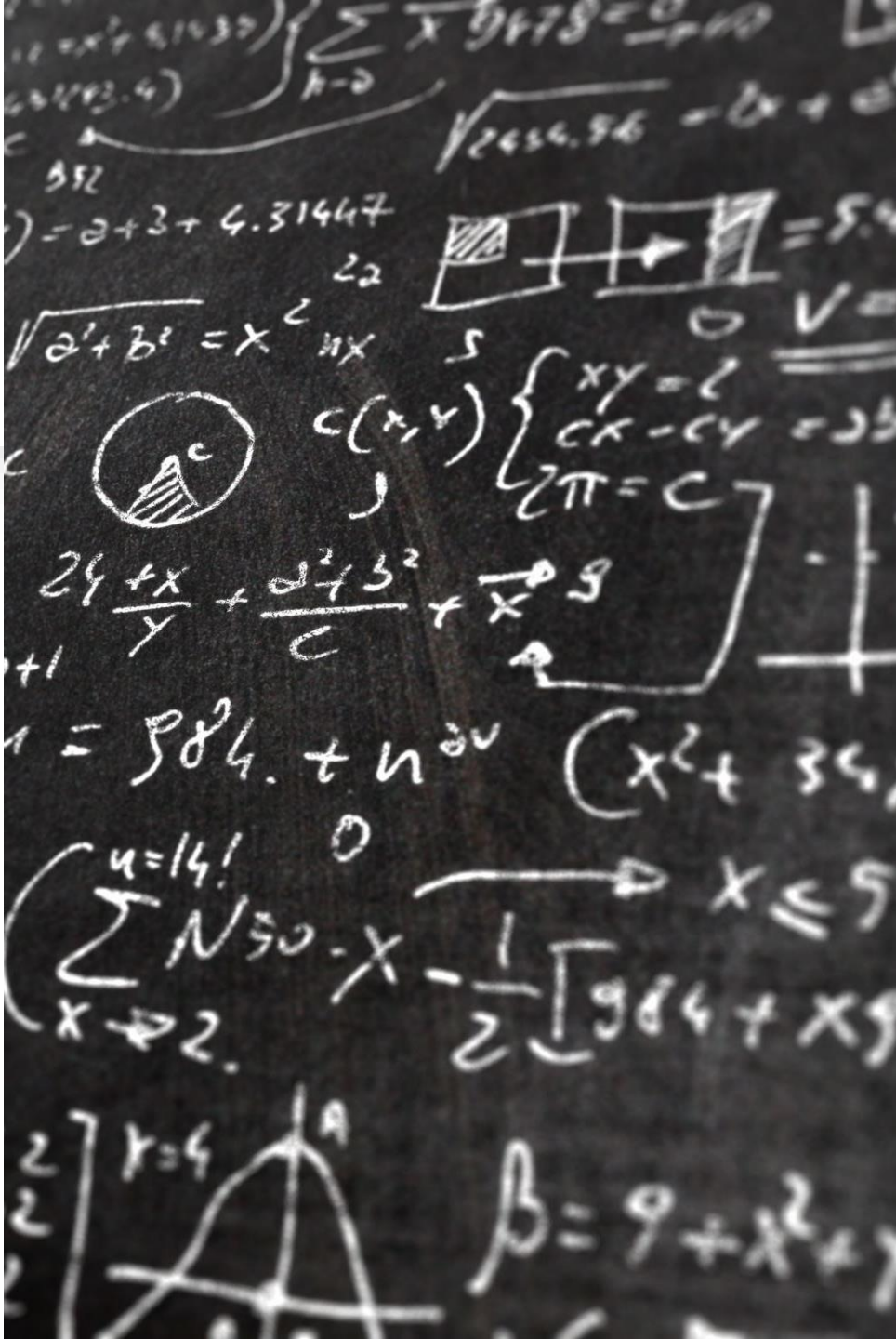
Comparison

SimCLR
Strong Discrimination

BYOL
Stability

MAE
Structural Learning

Conclusion

- SSL improves representation learning
- Handles small dataset challenges

- Final Strategy:
  SimCLR + BYOL + MAE